Reconstruction for an inverse scattering problem with a Kerr type nonlinearity

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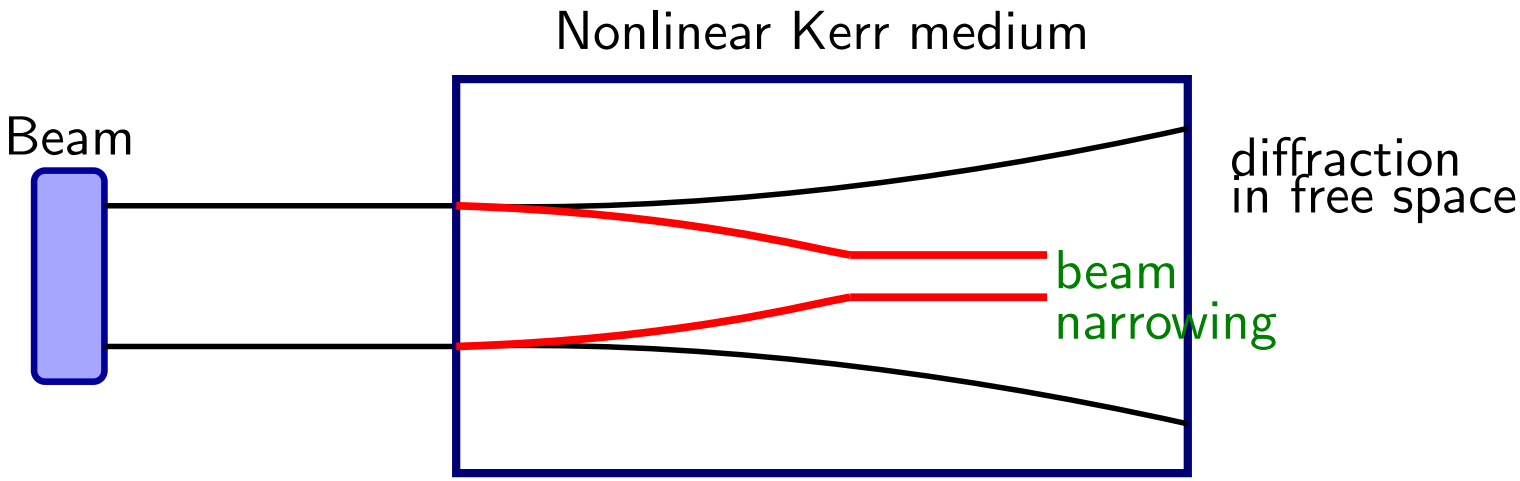








Physical motivation



In free space, diffraction makes the beam spread. In a Kerr medium, the nonlinear refractive index makes the beam narrow.

Kerr-type nonlinear Helmholtz equation

Let $q \in L^p_{\mathrm{comp}}(\mathbb{R}^n)$ with $p > \max\{2, n/2\}$.

$$\begin{cases} \Delta u + k^2\big(1 + q(x)|u|^2\big)u = 0, & x \in \mathbb{R}^n, \\ u = u_{\mathrm{inc}} + u_{\mathrm{sc}}, & \\ u_{\mathrm{inc}}(x) = \varepsilon e^{ik\theta\cdot x}, & \theta \in \mathbb{S}^{n-1},\ k > 0, \\ \lim\limits_{|x|\to\infty} |x|^{\frac{n-1}{2}} \left(\dfrac{x}{|x|}\cdot\nabla - ik\right) u_{\mathrm{sc}}(x) = 0. & \end{cases}$$

where $k > 0$ is the wavenumber, u_{inc} is the incident plane wave of amplitude $\varepsilon > 0$, and u_{sc} is the scattered field satisfying the Sommerfeld radiation condition.

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Scattering amplitude

Let $q \in L^p_{\mathrm{comp}}(\mathbb{R}^n)$ with $p > \max\{2, n/2\}$.

$$\begin{cases} \Delta u + k^2\left(1 + q(x)|u|^2\right)u = 0, & x \in \mathbb{R}^n, \\ u = u_{\mathrm{inc}} + u_{\mathrm{sc}}, & \\ u_{\mathrm{inc}}(x) = \varepsilon e^{ik\theta\cdot x}, & \theta \in \mathbb{S}^{n-1},\ k > 0, \\ \lim\limits_{|x|\to\infty} |x|^{\frac{n-1}{2}} \left(\dfrac{x}{|x|}\cdot\nabla - ik\right) u_{\mathrm{sc}}(x) = 0. & \end{cases}$$

the corresponding scattering amplitude

$$A(k, \hat{\theta}, \theta; \varepsilon) = \int_{\mathbb{R}^n} e^{-ik\hat{\theta}\cdot y} q(y)|u(y)|^2 u(y)dy.$$

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Theorem (Tyni, Pohjola, Liimatainen, Lassas, E-M.)

Let $q \in L^p(\mathbb{R}^n)$, with $p > \max\{2, n/2\}$, be compactly supported. Let $U \subset \mathbb{R}_+$ and $V, W \subset \mathbb{S}^{n-1}$. Define the set of accessible frequencies

$$M := \{\xi := k(\hat{\theta} - \theta) \mid k \in U, \theta \in V, \hat{\theta} \in W\} \subset \mathbb{R}^n.$$

Then the nonlinear scattering amplitude

$$A(k, \hat{\theta}, \theta; \varepsilon), \quad u_{\text{inc}} = \varepsilon e^{ik\theta \cdot x},$$

for all sufficiently small $\varepsilon > 0$, all $k \in U$, $\theta \in V$, and $\hat{\theta} \in W$, uniquely determines $\mathcal{F}(q)(\xi)$ for all $\xi \in M$. If M contains an open subset, then these scattering data uniquely determine q.

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Proof of the Theorem: Third Linearization

Let

$$v := D_\varepsilon u\big|_{\varepsilon=0} = e^{ik\theta\cdot x}.$$

The third-order linearization of the scattering amplitude gives

$$\begin{aligned} D_\varepsilon^3 A(k,\hat{\theta},\theta;\varepsilon)\Big|_{\varepsilon=0} &= 6\int_{\mathbb{R}^n} e^{-ik\hat{\theta}\cdot y}\, q(y)\,|v(y)|^2 v(y)\,dy \\ &= 6\int_{\mathbb{R}^n} q(y)\, e^{-ik(\hat{\theta}-\theta)\cdot y}\,dy \\ &= 6\,\mathcal{F}(q)\big(k(\hat{\theta}-\theta)\big) \\ &= 6\,\mathcal{F}(q)\big(\xi\big), \quad \xi = k(\hat{\theta}-\theta). \end{aligned}$$

Thus, we obtain

$$\boxed{\mathcal{F}(q)(\xi) = \frac{1}{6} D_\varepsilon^3 A(k,\hat{\theta},\theta;\varepsilon)\Big|_{\varepsilon=0}}$$

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Proof of the Theorem:

Recover the Fourier transform on M

Since

$$\xi = k(\hat{\theta} - \theta) \in M,$$

the scattering data determine

$$\boxed{\mathcal{F}(q)(\xi) \quad \text{for every } \xi \in M.}$$

Analytic continuation

Suppose that M contains a nonempty open set O.

Since q is compactly supported, the Paley-Wiener-Schwartz theorem implies that $\mathcal{F}(q)$ is real-analytic on $\mathbb{R}^n$.

Hence its values on O uniquely determine $\mathcal{F}(q)$ on all of $\mathbb{R}^n$.

$$\boxed{\mathcal{F}(q) \implies q}$$

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Corollary

The mapping $u_{\text{inc}} \mapsto A$ uniquely determines q in the following cases:

1. **Full data:** *if $U = (0, \infty)$ and $V = W = \mathbb{S}^{n-1}$, then $M = \mathbb{R}^n \setminus \{0\}$.*
2. **Backscattering:** *if $U = (k_1, k_2)$ for some $0 \leq k_1 < k_2 < \infty$ and $\hat{\theta} = -\theta \in \mathbb{S}^{n-1}$, then $M = \{\xi \in \mathbb{R}^n \mid 2k_1 < |\xi| < 2k_2\}$.*
3. **Fixed energy:** *if $k = k_0 \in (0, \infty)$ is fixed and $V = W = \mathbb{S}^{n-1}$, then $M = \{\xi \in \mathbb{R}^n \mid |\xi| \leq 2k_0\}$.*
4. **Fixed angle:** *if $\theta = \theta_0 \in \mathbb{S}^{n-1}$ (resp. $\hat{\theta} = \theta_0 \in \mathbb{S}^{n-1}$) is fixed, and $U = (k_1, k_2)$ for some $0 \leq k_1 < k_2 \leq \infty$, then M is a (truncated) cone. Thus, if U contains an open interval and W (resp. V) has a relatively open subset on the sphere, then M contains an open subset.*

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Lippmann-Schwinger formulation

We consider the nonlinear Helmholtz equation in two dimensions

$$\Delta u(x) + k^2\left(1 + q(x)\,|u(x)|^2\right)u(x) = 0, \qquad x \in \mathbb{R}^2,$$

The nonlinear Lippmann-Schwinger integral equation

$$u(x) = u_{\text{inc}}(x) + \int_{\mathbb{R}^2} G_k(x-y)\,k^2 q(y)\,|u(y)|^2 u(y)\,dy,$$

where in the two dimensional case the radiating fundamental solution of the Helmholtz operator is

$$G_k(x-y) = \frac{i}{4}H_0^{(1)}(k|x-y|).$$

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Lippmann-Schwinger formulation

We consider the nonlinear Helmholtz equation in two dimensions

$$\Delta u(x) + k^2\left(1 + q(x)\,|u(x)|^2\right)u(x) = 0, \qquad x \in \mathbb{R}^2,$$

The nonlinear Lippmann-Schwinger integral equation

$$u(x) = u_{\mathrm{inc}}(x) + \int_B G_k(x-y)\,k^2 q(y)\,|u(y)|^2 u(y)\,dy,$$

Since q is supported in the ball B, the domain of integration reduces to B.

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Newton method

We define the nonlinear operator $\mathcal{T} : L^\infty(B_{2R}) \to L^\infty(B_{2R})$ by

$$\mathcal{T}(u)(x) = u(x) - u_{\text{inc}}(x) - \int_B G_k(x-y)\, k^2 q(y)\, |u(y)|^2 u(y)\, dy,$$

The nonlinear system $\mathcal{T}(u) = 0$ is solved using Newton's method.
We will solve the system iteratively as

$$u^{n+1} = u^n + \delta u^n, \quad u_0 = u_{\text{inc}},$$

where the Newton correction δu^n satisfies

$$\mathcal{T}'(u^n)[\delta u^n] = -\mathcal{T}(u^n).$$

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Vainikko method

At iteration n, the correction δu^n satisfies the real-linearized integral equation

$$\delta u^n(x) - \int_B G_k(x-y)\, k^2 q(y) \Big(2|u^n(y)|^2 \delta u^n(y) + (u^n(y))^2 \overline{\delta u^n(y)} \Big)\, dy = -\mathcal{T}(u^n)(x).$$

Vainikko method

Using the convolution theorem, this becomes

$$\delta u^n(x) - \mathcal{F}^{-1}\Big[\mathcal{F}(G_k)(\xi)\,\mathcal{F}\Big(k^2 q\big(2|u^n|^2\delta u^n + (u^n)^2\overline{\delta u^n}\big)\Big)(\xi)\Big](x) = -\mathcal{T}(u^n)(x).$$

After discretization, the Fourier coefficients of G_k are computed explicitly using Vainikko's formulas.

G. Vainikko, *Fast solvers of the Lippmann-Schwinger equation*, 2000, p. 16.

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Discretization

Uniform Cartesian grid with

$$2N \times 2N = 2^M \times 2^M$$

grid points,

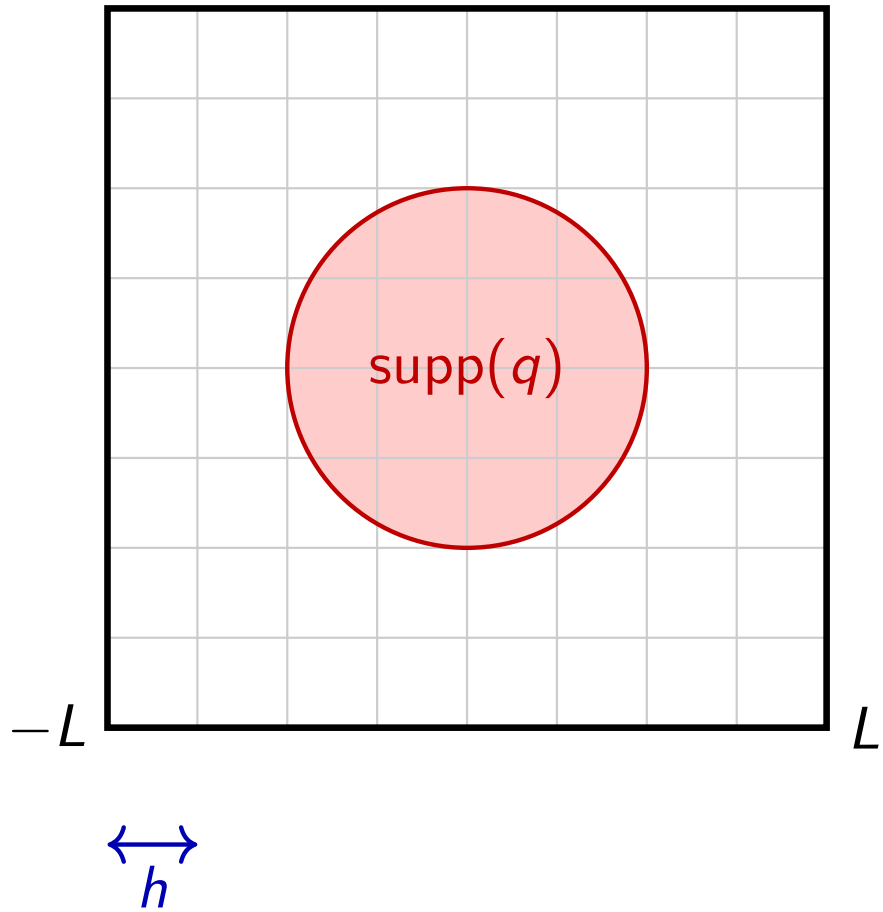
$$h = \frac{2L}{2N-1},$$

and the grid points are

$$x_{ij} = \big(-L + ih,\ -L + jh\big),$$

where

$$0 \le i, j \le 2N - 1 = 2^M - 1.$$



$$\Omega = [-L, L]^2, \qquad L = 2R$$

Matrix form

Denoting the (complex-valued) discretized linearized equation by

$$B[u^n](\delta u^n) = Y, \qquad \delta u^n \in \mathbb{C}^N, \quad Y \in \mathbb{C}^N,$$

Convert the complex system into a real one $\mathbb{C}^N \cong \mathbb{R}^{2N}$.
Let

$$\delta u^n = V_R + iV_I, \qquad V_R, V_I \in \mathbb{R}^N, \quad \text{and} \quad V = \begin{pmatrix} V_R \\ V_I \end{pmatrix} \in \mathbb{R}^{2N}.$$

Similarly, we define

$$Y = Y_R + iY_I \qquad Y_R, Y_I \in \mathbb{R}^N, \quad \text{and} \quad \widetilde{Y} = \begin{pmatrix} Y_R \\ Y_I \end{pmatrix} \in \mathbb{R}^{2N}.$$

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Real-linear Matrix

Define the real-linear operator $\widetilde{B}[u^n] : \mathbb{R}^{2N} \to \mathbb{R}^{2N}$ by

$$\widetilde{B}[u^n](V) = \begin{pmatrix} \mathrm{Re}\left(B[u^n](V_R + iV_I)\right) \\ \mathrm{Im}\left(B[u^n](V_R + iV_I)\right) \end{pmatrix}.$$

The original complex system is thus equivalent to the real system

$$\widetilde{B}[u^n](V) = \widetilde{Y},$$

which is linear over $\mathbb{R}$ and has dimension $2N$.
Thus the use of GMRES solver.

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Backscattering:

In the case of backscattering $\hat{\theta} = -\theta$, and the third linearization gives

$$D_\varepsilon^3 A(k, -\theta, \theta; \varepsilon)\big|_{\varepsilon=0} = 6\,\mathcal{F}(q)(-2k\theta).$$

The sampled frequencies are

$$\xi = -2k\theta,$$

and for isotropic resolution, we sample the frequency points ξ in polar coordinates, that is,

$$\xi_r = -2k_\ell\theta_j = -2k_\ell(\cos\varphi_j,\ \sin\varphi_j) = (\xi_{r,x}, \xi_{r,y}).$$

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Fixed angle scattering, $\hat{\theta}$ fixed:

We fix the observation direction $\hat{\theta} = \hat{\theta}_0$. Then

$$D_\varepsilon^3 A(k, \hat{\theta}_0, \theta; \varepsilon)\big|_{\varepsilon=0} = 6\,\mathcal{F}(q)\big(k(\hat{\theta}_0 - \theta)\big).$$

The frequencies are given by

$$\xi = k(\hat{\theta}_0 - \theta),$$

and we sample the frequency points ξ in polar coordinates as

$$\xi_r = k_\ell(\hat{\theta}_0 - \theta_j) = k_\ell(\hat{\theta}_{0,1} - \cos\varphi_j,\ \hat{\theta}_{0,2} - \sin\varphi_j) = (\xi_{r,x}, \xi_{r,y}),$$

Fixed energy scattering, k is fixed:

When the frequency $k = k_0$ is fixed, one obtains

$$D_\varepsilon^3 A(k_0, \hat{\theta}, \theta; \varepsilon)\big|_{\varepsilon=0} = 6\,\mathcal{F}(q)\big(k_0(\hat{\theta} - \theta)\big).$$

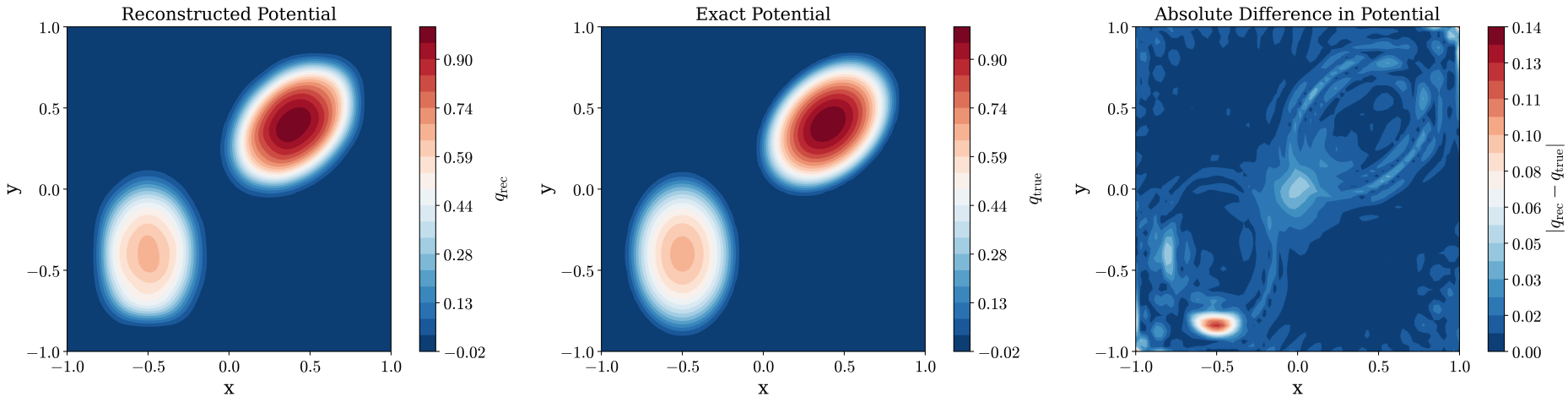
Here the frequency points

$$\xi = k_0(\hat{\theta} - \theta),$$

cover a closed disk $\overline{B_{2k_0}}$, and we sample the frequency points ξ in polar coordinates

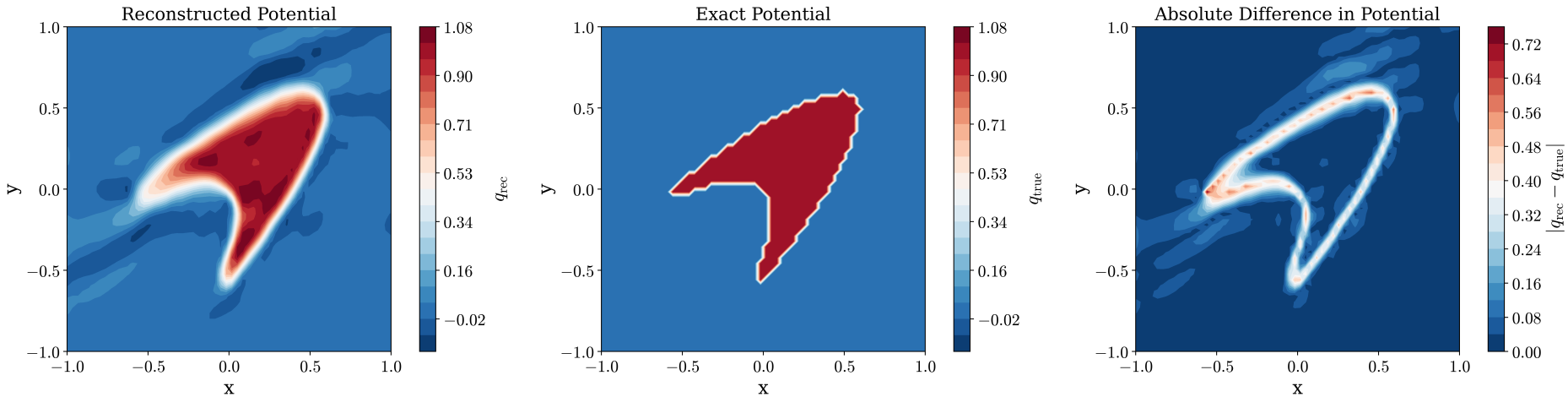
$$\xi_r = k_0(\hat{\theta}_j - \theta_j) = k_0(\cos\phi_j - \cos\varphi_j,\ \sin\phi_j - \sin\varphi_j) = (\xi_{r,x}, \xi_{r,y}),$$

Example: two elliptic bumps



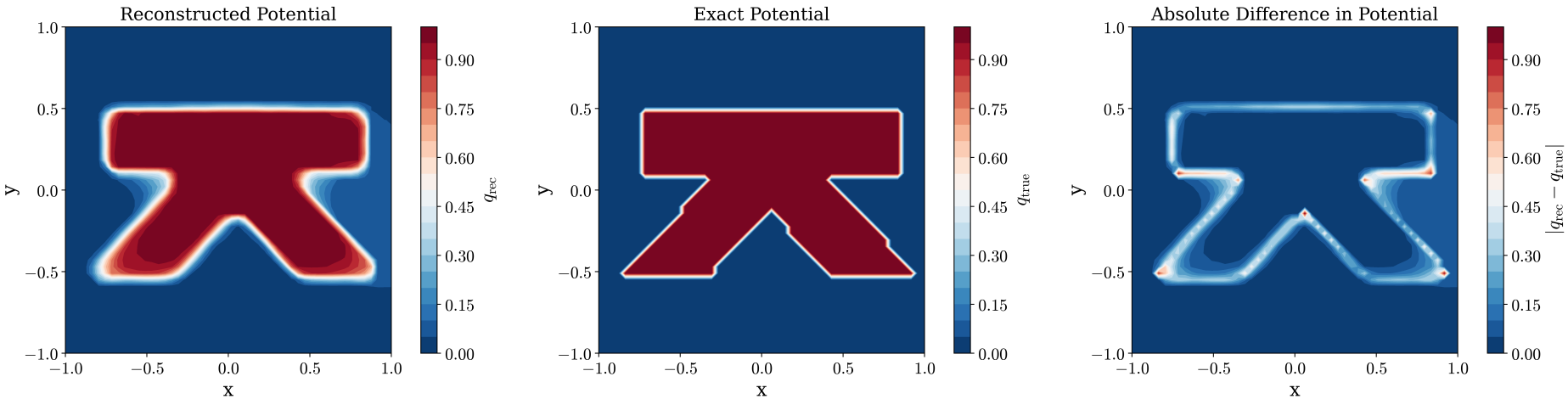
Two elliptic bumps reconstruction

Example: kite-shape



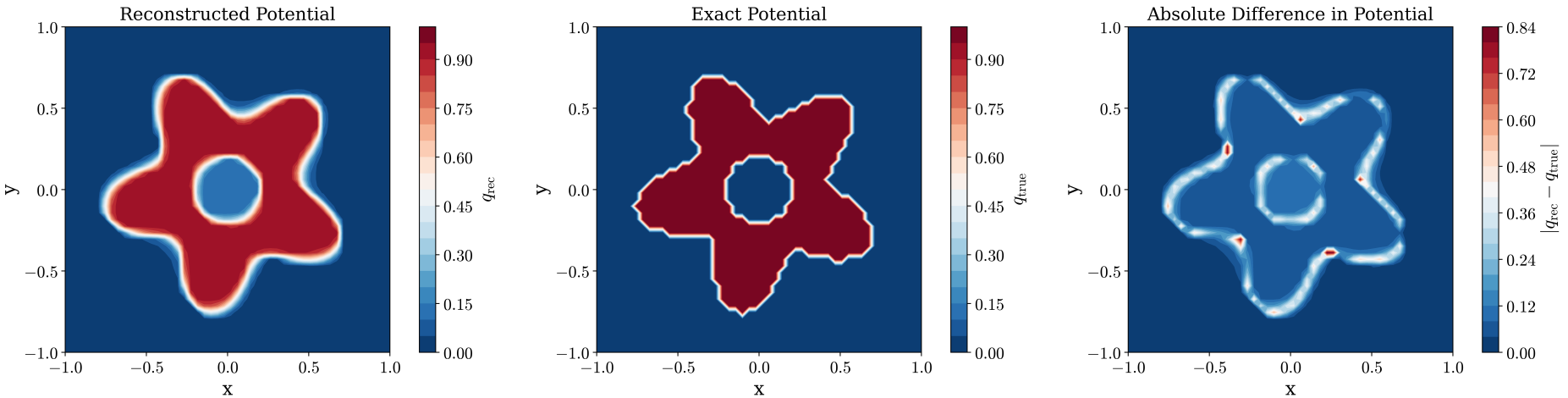
Kite-shape reconstruction

Example: K-shape



K-shape reconstruction

Example: flower-shape



Flower-shape reconstruction


Thank you for your attention!
Khaoula El Maddah